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Pytorch TensorFlow

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End-of-chapter quiz

1. What is the order of the language modeling pipeline?

- [ ] First, the model, which handles text and returns raw predictions. The tokenizer then makes sense of these predictions and converts them back to text when needed.
- [ ] First, the tokenizer, which handles text and returns IDs. The model handles these IDs and outputs a prediction, which can be some text.
- [x] The tokenizer handles text and returns IDs. The model handles these IDs and outputs a prediction. The tokenizer can then be used once again to convert these predictions back to some text.

Correct! The tokenizer can be used for both tokenizing and de-tokenizing.

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2. How many dimensions does the tensor output by the base Transformer model have, and what are they?

- [ ] 2: The sequence length and the batch size
- [ ] 2: The sequence length and the hidden size
- [x] 3: The sequence length, the batch size, and the hidden size

Correct! Nicely done!

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3. Which of the following is an example of subword tokenization?

- [x] WordPiece

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Correct! Yes, that's one example of subword tokenization!

- [ ] Character-based tokenization
- [ ] Splitting on whitespace and punctuation
- [x] BPE

Correct! Yes, that's one example of subword tokenization!

- [x] Unigram

Correct! Yes, that's one example of subword tokenization!

- [ ] None of the above

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4. What is a model head?

- [ ] A component of the base Transformer network that redirects tensors to their correct layers
- [ ] Also known as the self-attention mechanism, it adapts the representation of a token according to the other tokens of the sequence
- [x] An additional component, usually made up of one or a few layers, to convert the transformer predictions to a task-specific output

Correct! That's right. Adaptation heads, also known simply as heads, come up in different forms: language modeling heads, question answering heads, sequence classification heads...

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5. What is an AutoModel?

- [ ] A model that automatically trains on your data
- [x] An object that returns the correct architecture based on the checkpoint

Correct! Exactly: the `AutoModel` only needs to know the checkpoint from which to initialize to return the correct architecture.

- [ ] A model that automatically detects the language used for its inputs to load the correct weights

Submit You got all the answers!

6. What are the techniques to be aware of when batching sequences of different lengths together?

- [x] Truncating

Correct! Yes, truncation is a correct way of evening out sequences so that they fit in a rectangular shape. Is it the only one, though?

- [ ] Returning tensors
- [x] Padding

Correct! Yes, padding is a correct way of evening out sequences so that they fit in a rectangular shape. Is it the only one, though?

- [x] Attention masking

Correct! Absolutely! Attention masks are of prime importance when handling sequences of different lengths. That's not the only technique to be aware of, however.

Submit You got all the answers!

7. What is the point of applying a SoftMax function to the logits output by a sequence classification model?

- [ ] It softens the logits so that they're more reliable.
- [ ] It applies a lower and upper bound so that they're understandable.
- [x] The total sum of the output is then 1, resulting in a possible probabilistic interpretation.

Correct! Correct! That's not the only reason we use a SoftMax function, though.

Submit You didn't select all the correct answers, there's more!

8. What method is most of the tokenizer API centered around?

- [ ] `encode`, as it can encode text into IDs and IDs into predictions
- [x] Calling the tokenizer object directly.

Correct! Exactly! The `__call__` method of the to
can handle pretty much anything. It is also the method used to retrieve predictions from a

model.

- [ ] pad
- [ ] tokenize

Submit You got all the answers!

9. What does the result variable contain in this code sample?

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")
result = tokenizer.tokenize("Hello!")
```

- [x] A list of strings, each string being a token

Correct! Absolutely! Convert this to IDs, and send them to a model!

- [ ] A list of IDs
- [ ] A string containing all of the tokens

Submit You got all the answers!

10. Is there something wrong with the following code?

```
from transformers import AutoTokenizer, AutoModel

tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")
model = AutoModel.from_pretrained("gpt2")

encoded = tokenizer("Hey!", return_tensors="pt")
result = model(**encoded)
```

- [ ] No, it seems correct.
- [x] The tokenizer and model should always be from the same checkpoint.

Correct! Right!

- [ ] It's good practice to pad and truncate with the tokenizer as every input is a batch.

Submit You got all the answers!

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